

Protecting Mining Sector Critical Technologies: Cyber Threats and Resilience Strategies

Creative
thinkers
made here.

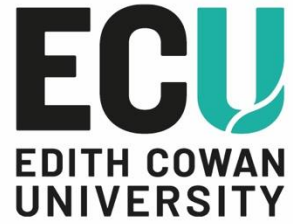
Speakers



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Cybersecurity
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Edith Cowan
University



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at Edith Cowan
University



What is this workshop meant for

What it covers for (Structure for all 3 Sessions)

Why we are suitable to deliver this workshop?

Creative
thinkers
made here.

Mining Sector Cybersecurity



Mining Sector Security

“Refers to the protection and resilience of critical physical and IT/OT-integrated assets supporting mining operations, such as exploration, extraction, processing, logistics, and monitoring systems, whose disruption could severely impact Australia’s economy, resource supply chains, and national security.”

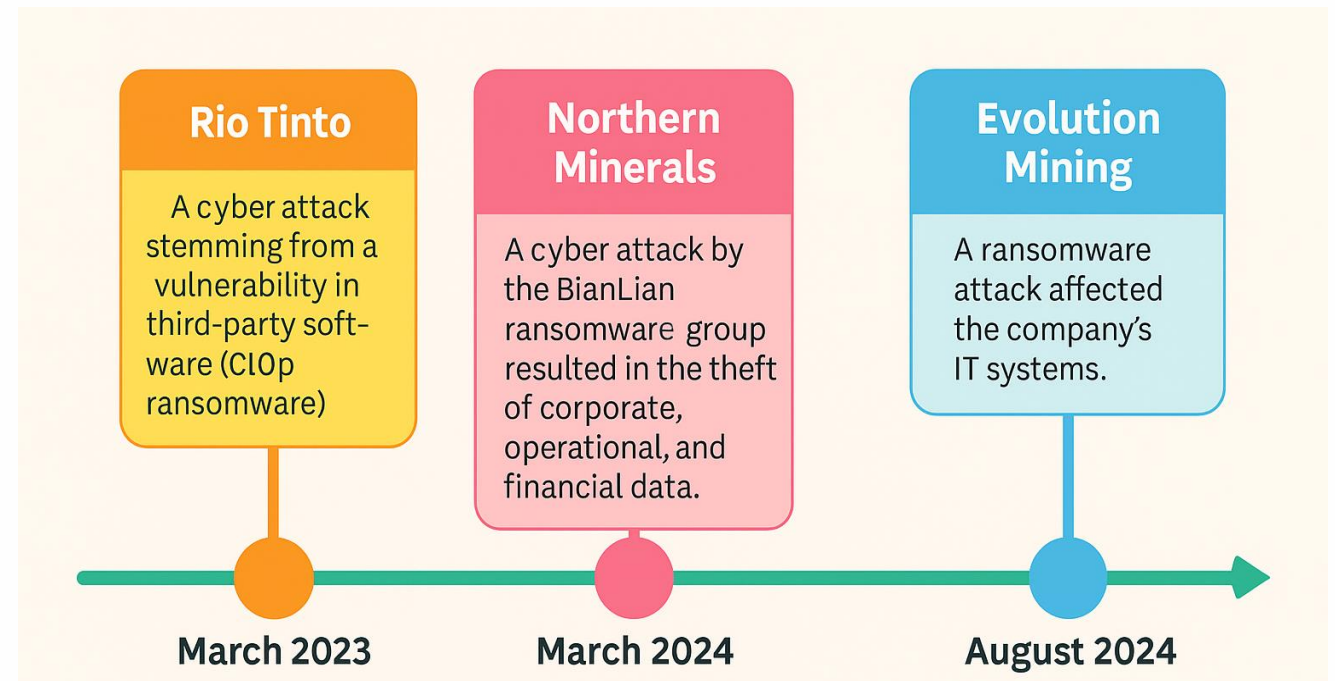
What do you think are most critical systems and components?

Mentimeter (Asks and then display word cloud) (Small Quiz here)



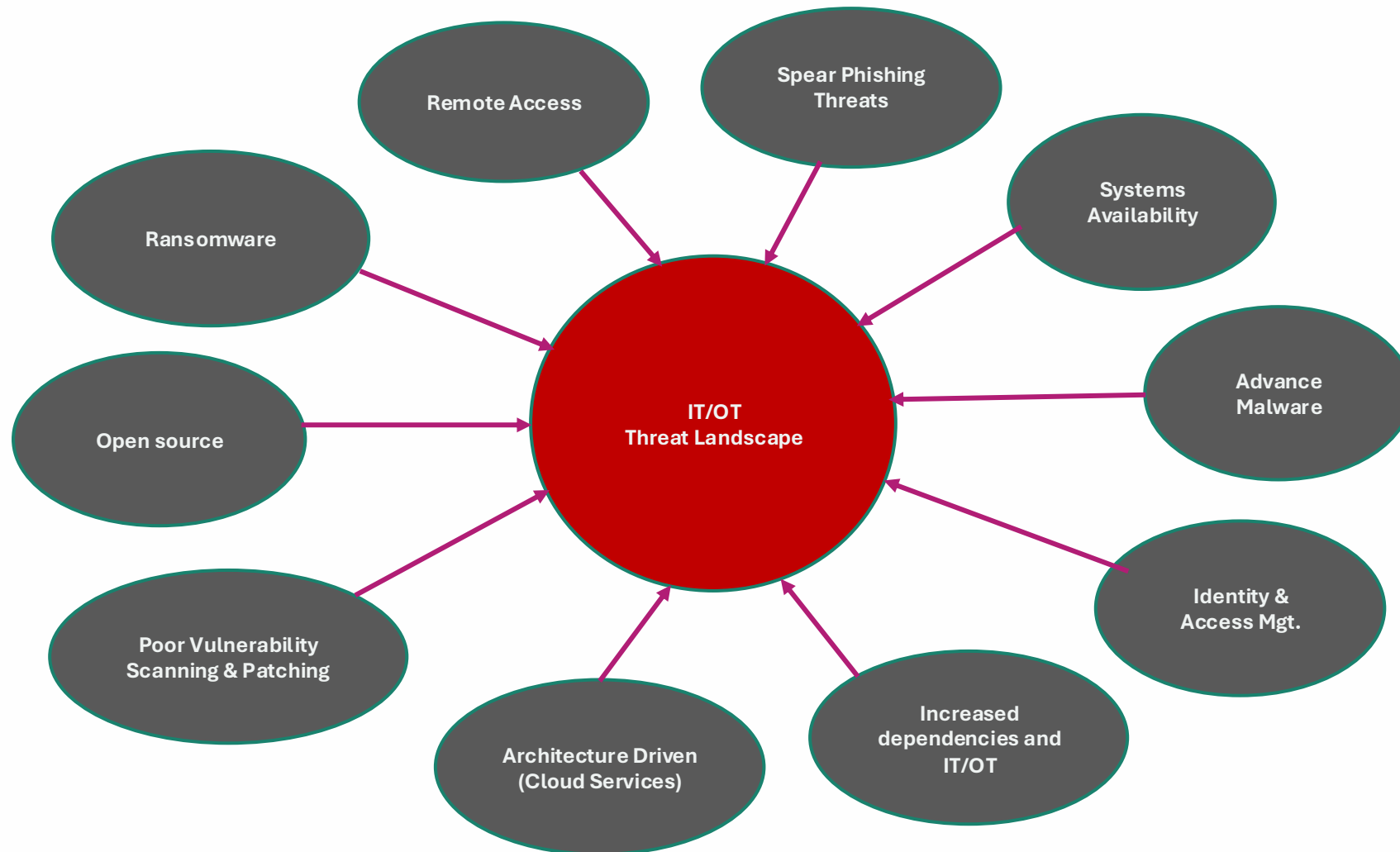
Cyber Attacks in Mining Sector

- Mining sector covers 50% of the Australia's export estimated \$450 Billions.
- Globally reported 65 cyber attacks out of which 5 Attacks were directly on Australia's critical Mining sector.
- Darktrace reports that 40% of Mining Equipment, Technology, and Services (METS) businesses in Australia have experienced cyber attacks in the past year.
- There is a lot of under reporting.

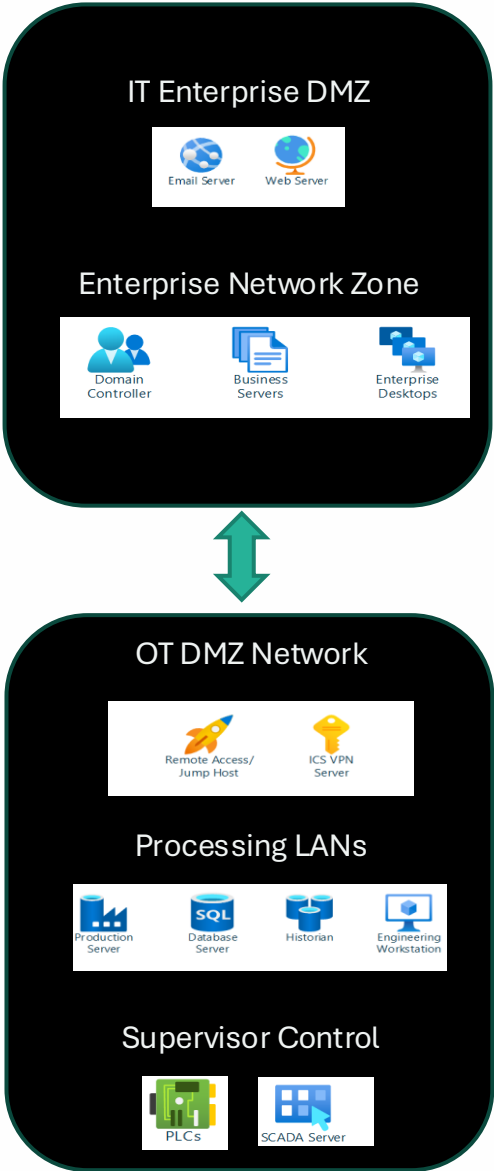
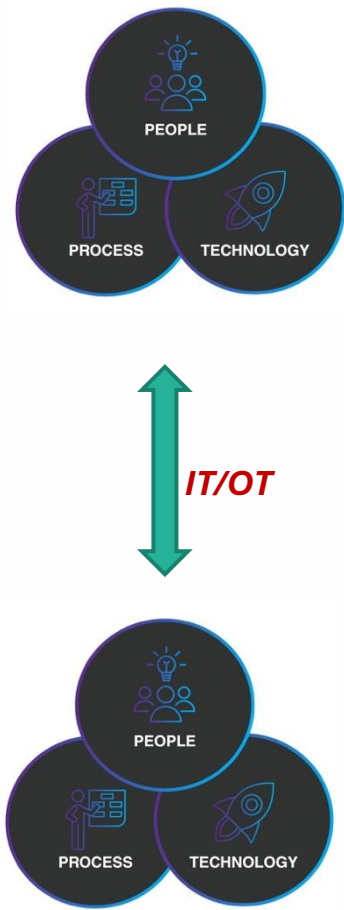


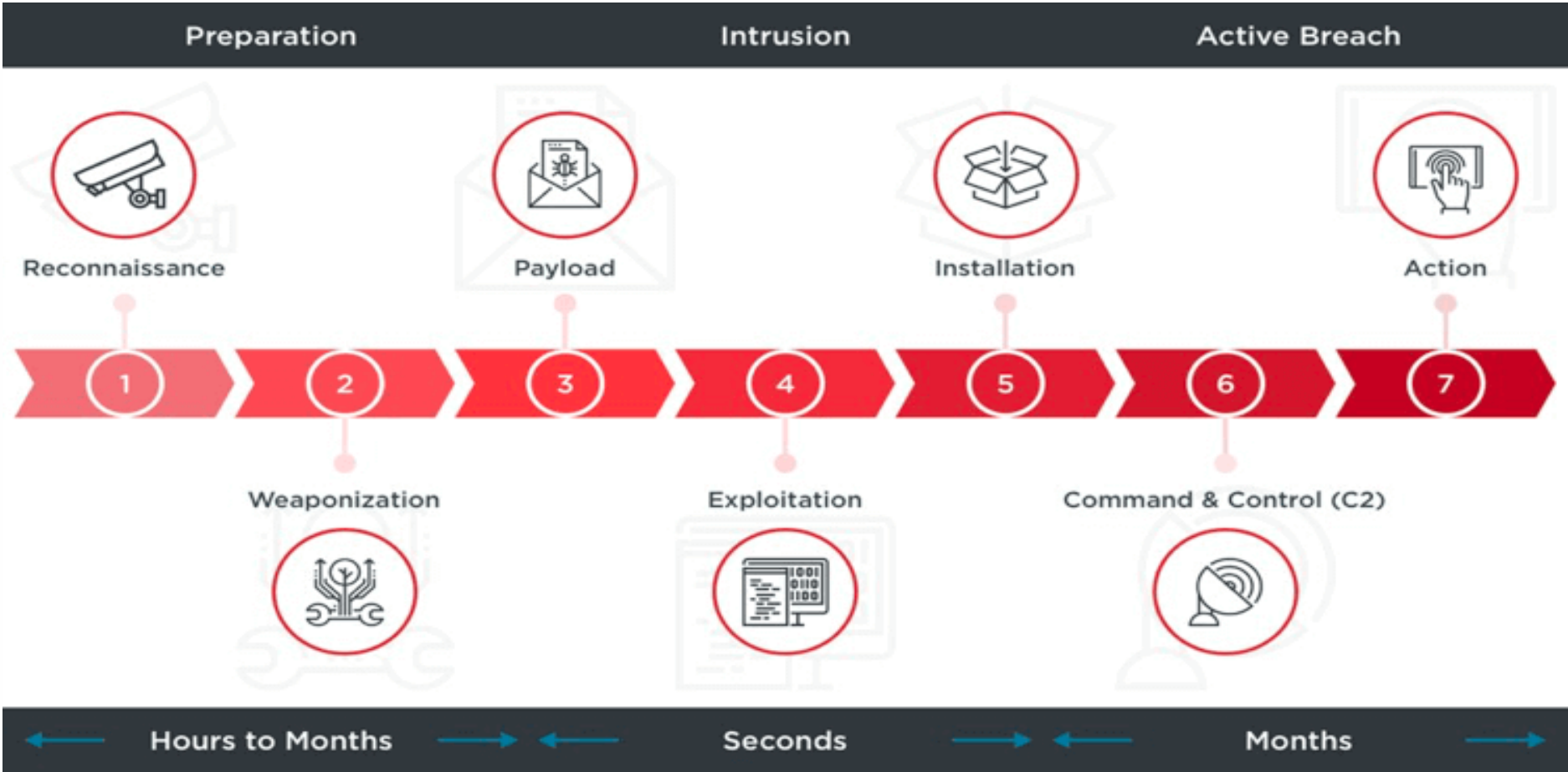
Sources, SecurityHQ, Group IB

Increasing Threat Landscape



Convergence of IT/OT and Challenges







Summary of the Session

What we learnt

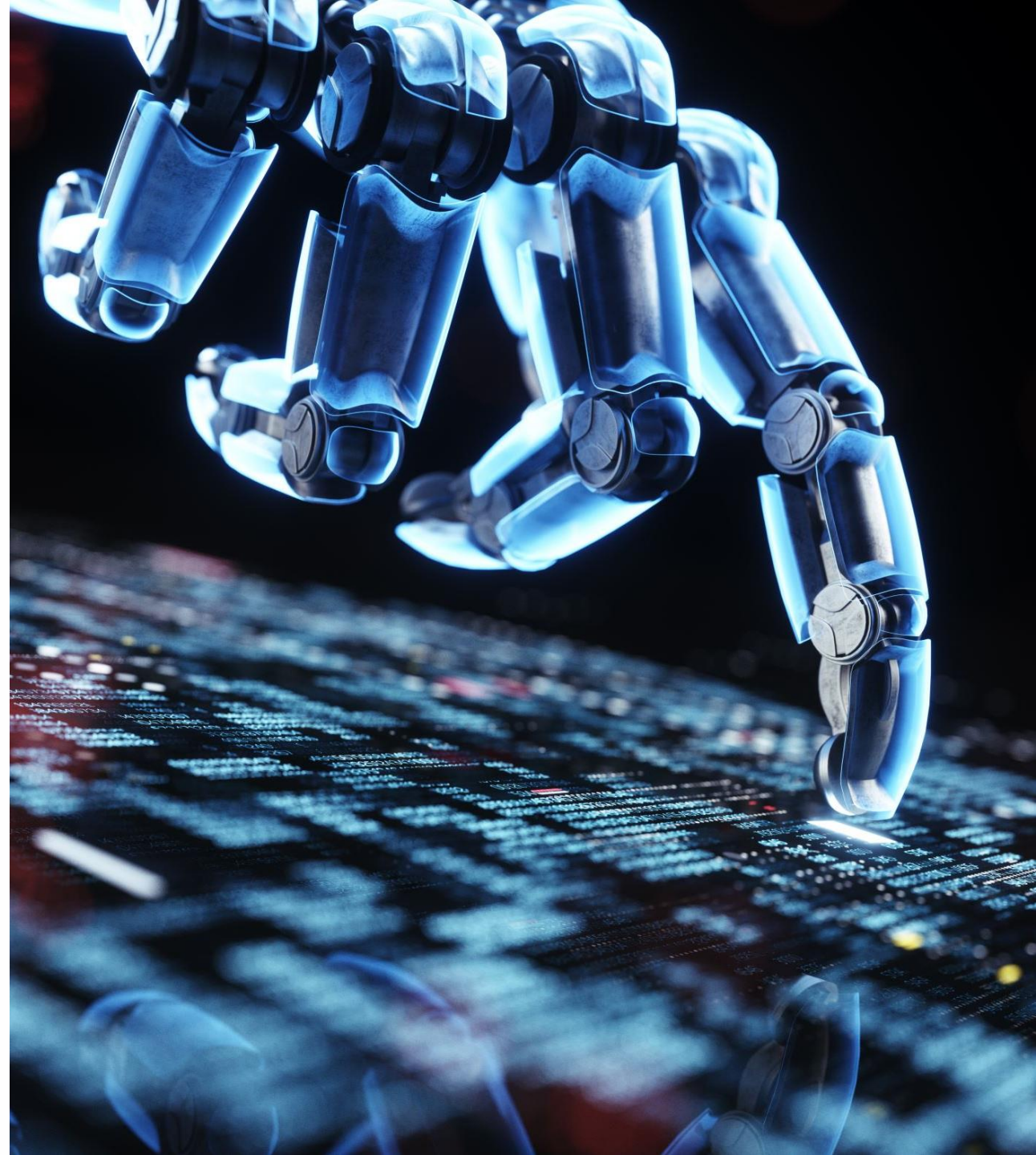
What's next

**Creative
thinkers
made here.**

SESSION- II

Walk through a cyber attack

Creative
thinkers
made here.



Attack Scenario 1- Iron Ore Logistics Operations- Train Supply Chain

Mine Site Operations



Iron Ore Freight Transport

Terminal Operations

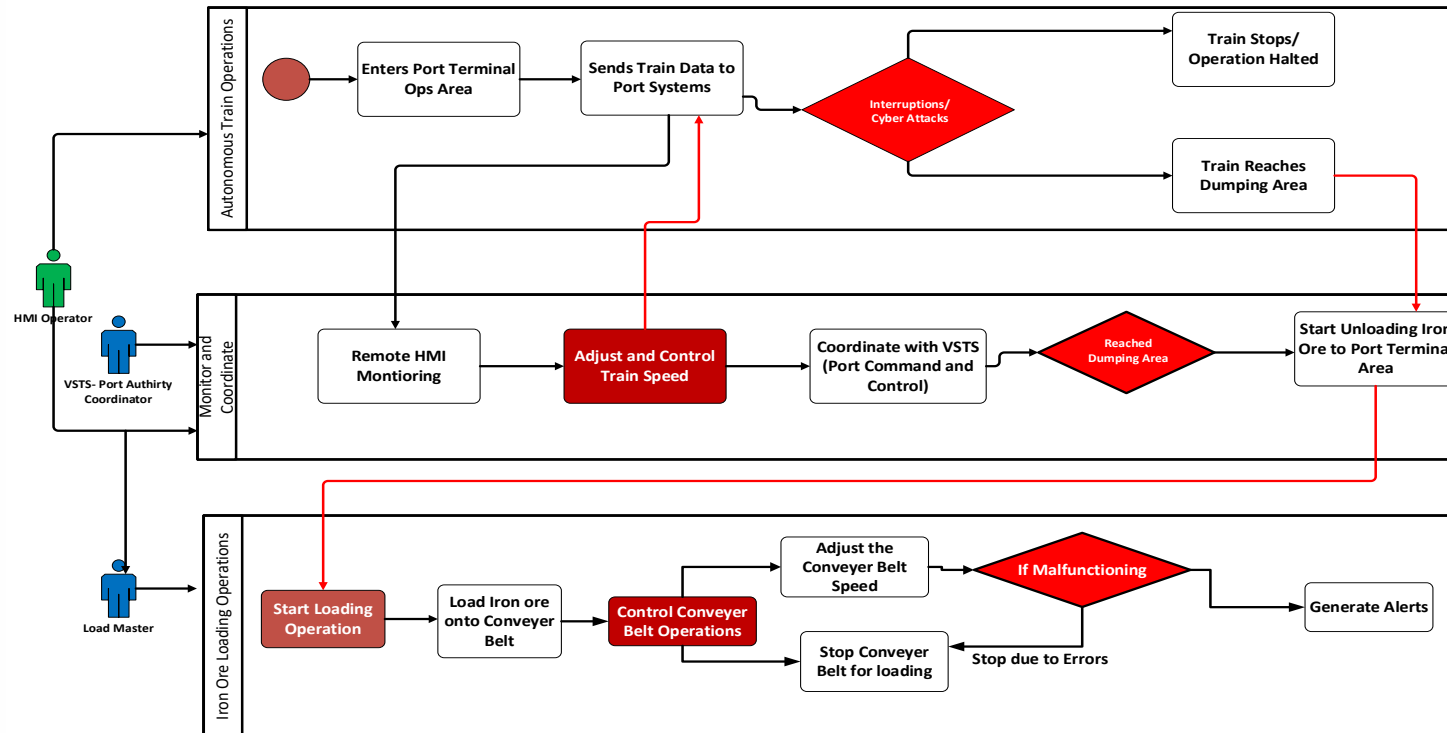


Scenario Overview

Company XYZ, a major mining operator, relies on integrated logistics for iron ore transport, from mine site to rail freight and finally port loading. A cyberattack disrupts OT systems, originated through IT systems, targeting scheduling, and SCADA operations, causing significant supply chain delays and operational downtime.

We will simulate Attackers tactics and Defenders strategies using cyber kill chain.

Scenario 1- Iron Ore Logistics Operations- Business processes



Iron Ore Logistics Operations Sabotage- IOCs, Attacker Tactic and Defensive Actions



Reconnaissance

Indicators of Compromise (IoC)

- IDs Nodes Probing from Unknown Sources
- Recon Activity Detected in Logs
- Network Mapping Scans Detected
- Noisy Parameters Scans
- Suspicious Activity from IT→ OT
- Unauthorized Queries towards IT/OT
- Employee Endpoint, Unsensual Activity

Example Attackers Tactics

- External Exposure Points
- Conduct Open-Source Intelligence
- Discover IT Systems and Internal Network Structure
- Identify External Exposure Points
- Map Interconnections Between IT and OT
- Harvest Employee Credentials

Tactical Decisions Choices

- 1. Block Suspicious Network Traffic
- 2. Isolate OT Segment Connected to IT
- 3. Sanitize IT and OT systems
- 4. Launch immediate Threat Hunt

QR Code to Scan

What Decisions you will take: Your Defensive Tactics (live Survey link)

SCAN QR CODE – Tactical Defense Decisions



Instructions:

- We will display Attackers Tactics, Indicators of Compromise and then you will choose most suitable choice(s) for defending against Attackers tactics.
- You can form a team or do it individually
- After scanning QR code you should be able to see each cyber kill chain stage.

Ahmed please put a screenshot here how the interface will look like (below)

QR CODE

Iron Ore Logistics Operations Sabotage- IOC's, Attacker Tactic and Defensive Actions



Weaponisation

IoCs

Unusual OT Protocol Traffic

Anomalous Network Behaviour

Unusual File Types Detected

Connections to suspicious servers

Repeated access to exposed endpoints

Unauthorized access attempts

Lateral movement via known admin

Attackers Tactics

Payload Development for OT Protocols

Crafting Supply Chain Exploits

Malware Wrapping in Legitimate Documents

Identify External Exposure Points

Custom Exploit Kits for HMI/SCADA Software

Dual-Use Tool Configuration

Tactical Decisions Choices

1. Malware Sandboxing

2. Block & Monitor Known Malicious Infrastructure

3. Isolate suspicious endpoints

4. Lock Out a Privileged Account Mid-Shift

What Decisions you will take: Your Defensive Tactics (live Survey link)

Iron Ore Logistics Operations Sabotage- Indicators of Compromise



Delivery

IoCs

Email Gateway Alerts for Suspicious Attachments

Unrecognized USB Device Activity

Unauthorized Remote Access Attempts

Unexpected File Uploads Detected

Repeated access to exposed endpoints

Outbound Traffic to Unfamiliar IPs After File Execution

Attackers Tactics

Phishing Emails to IT Systems and Employees Endpoints

Dropping Infected USB Devices at Remote Sites

Abusing Remote Access Tools or VPNs

Malicious File Uploads to Shared Portals

Leveraging Compromised Third-Party Software

Tactical Decisions Choices

1. Temporarily revoke access for external contractors and OEM vendors

2. Quarantine Affected Operational Assets

3. Proactively scan document repositories, shared drives, and endpoint telemetry

4. Trigger Conditional Access Controls

Iron Ore Logistics Operations Sabotage- Indicators of Compromise



Exploitation

Attackers Tactics

- Exploit Unpatched SCADA/HMI Vulnerabilities
- Execute Macros or Scripted Payloads
- Exploit Default or Weak Credentials in OT Devices
- Trigger Remote Code Execution in Load Scheduling Systems

IoCs

- Abnormal Remote Access Attempts to SCADA/HMI
- Endpoint logs suspicious Execution of PowerShell Scripts
- Authentication Failures or Unusual Logins to OT Devices
- Unexpected Configuration Changes in Load Scheduling Systems

Tactical Decisions Choices

- 1. Isolate or Quarantine Compromised OT Systems Immediately
- 2. deploy Incident Response Team to Conduct Live Forensics
- 3. Rollback Affected Load Scheduling/Weighbridge Configurations
- 4. Maintain Current Setup Under Monitoring

Iron Ore Logistics Operations Sabotage- Attackers' Tactics



Installation

IoCs

Unexpected Binary Files in Load Scheduling Terminals

Unauthorized Modifications in Weighbridge PLC Firmware

Abnormal Outbound Traffic from HMI to External IPs

Irregular Vendor Update Events Logged

Attackers Tactics

Deploy Malware into Load Scheduling Terminals

Install Backdoors on Weighbridge PLC Controllers

Establish Covert Channels via Compromised HMI Interfaces for Data Exfiltration

inject Malicious Code into Third-Party Update Mechanisms

Tactical Decisions Choices

1. Suspend all remote operations

2. Conduct Integrity Verification of PLCs and HMIs

3. Initiate Controlled Restoration of Affected Terminals from Clean Images

4. AirGap Entire Affected OT Segment from Network

Iron Ore Logistics Operations Sabotage- Attackers' Tactics



Command & Control

IoCs

Frequent Outbound Connections to Unknown IPs

Detection of Persistence Mechanisms

Unusual Remote Access Sessions Originating Outside Business Hours

Encrypted Traffic from OT terminals to External Networks

Attackers Tactics

Establish Encrypted Remote Access via Compromised Endpoints

Maintain Persistence via Scheduled Scripts

Use Beacons from Compromised PLCs to External Hosts

Leverage Third-Party Remote Management Channels

Tactical Decisions Choices

1. Force Terminate All Active Remote Sessions

2. Conduct Deep Packet Inspection (DPI) on OT Outbound Traffic

3. Disable Third-Party Remote Access Channels Until Validated

4. Migrate Affected Systems to an Isolated Forensic Zone

Iron Ore Logistics Operations Sabotage- Attackers' Tactics

Actions & Objectives

Attackers Tactics

Sabotage Logistics Operations by attacking IT and OT Systems

**Manipulate Dispatch Data in Terminal IT System.
Cargo misdirection's, impact supply chain**

Trigger Emergency Shutdown Protocols via SCADA

Exfiltrate and Leak Sensitive Operational Data

Tactical Decisions Choices

**1. Decide Whether to
Shut Down Logistics
Operation Temporarily**

**2. Communicate Incident
Severity to Executives and
Regulators**

**3. Disclose Incident to
Shareholders, Partners, or
Public?**

**4. Launch Strategic
Review of OT
Cybersecurity Resilience**

De- Briefing Attack Scenario 1

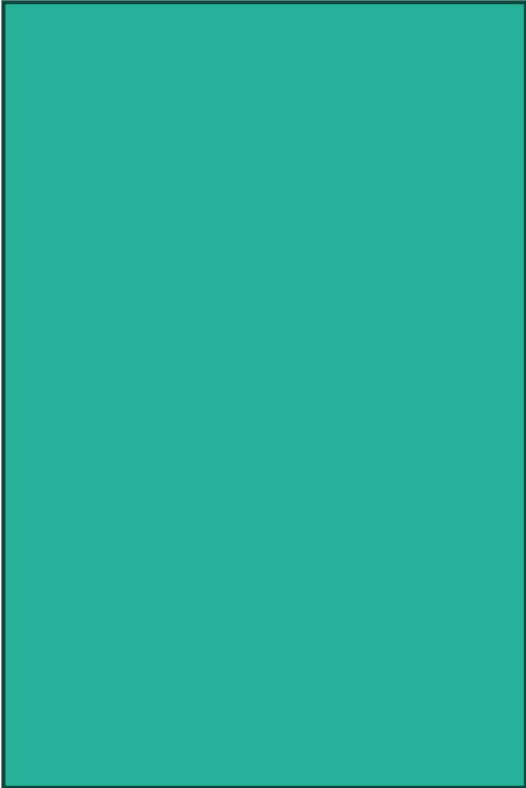
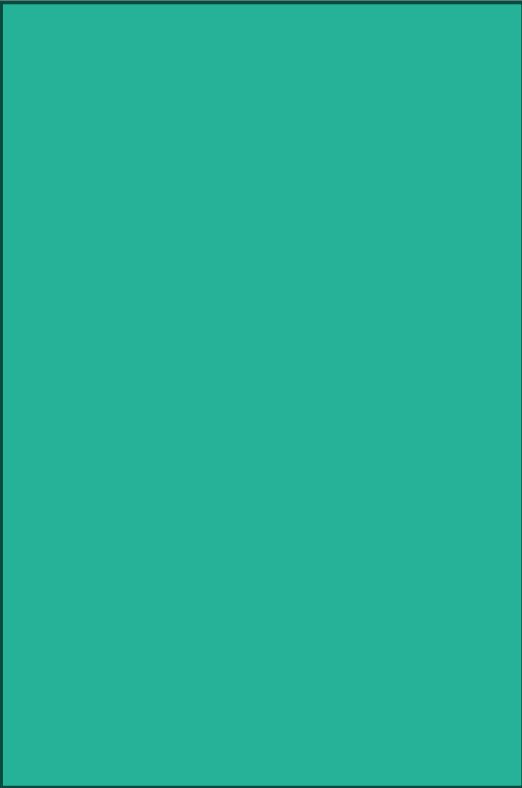
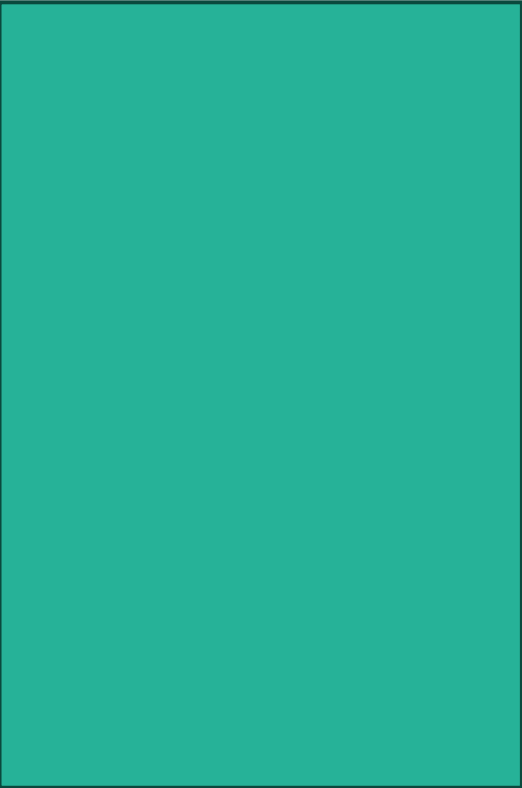
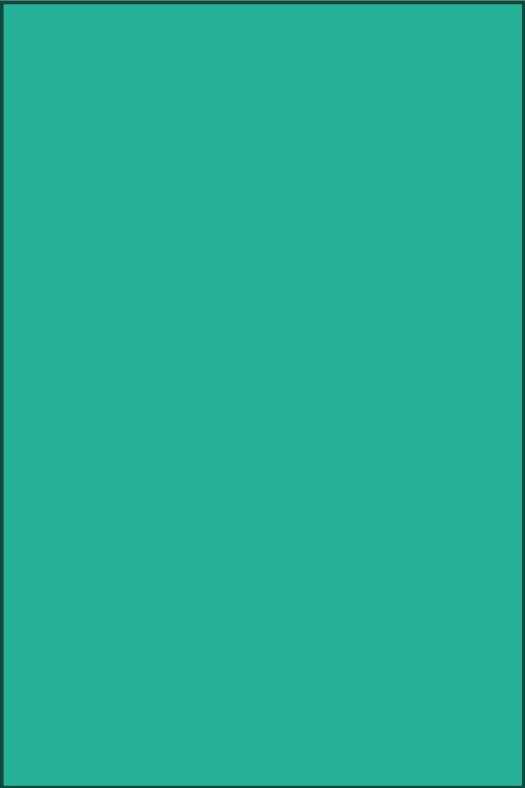
Iron Ore Logistics Operations Sabotage- Debrief

Reconnaissance

Weaponisation

Delivery

Exploitation



Iron Ore Logistics Operations Sabotage- Debrief

Installation

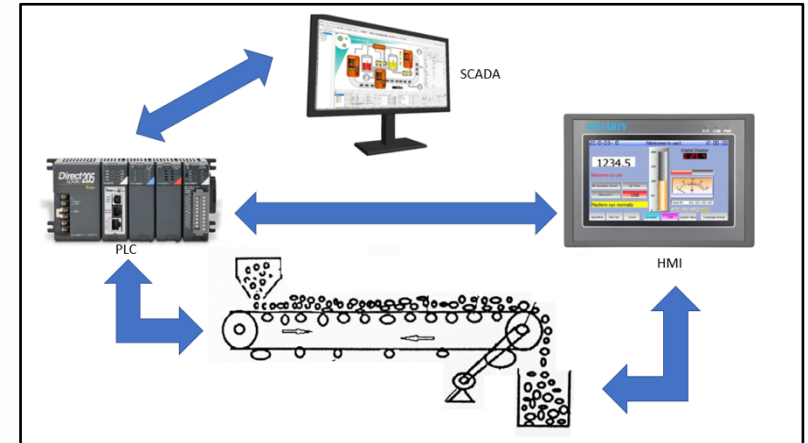
Command & Control

Actions &
Objectives

Mining Sector- Attack Scenarios

Scenario 2- Mining Material Loading and Unloading –Conveyer Belt Operations

In Scenario 2, an insider at Company XYZ manipulates mining conveyor belt operations by tampering with PLC logic and OT controls. The objective is to disrupt automated iron ore loading, causing operational delays, equipment faults, and potential safety risks. Ask ChatGPT



Scenario 2- Mining Materials Conveyor Belt Ops- Attacker Tactics

Reconnaissance

Attackers Tactics

Access Internal Network
Information

Review Employees Shift Schedules
and Access Logs

Identify Supply Chain and Vendors

Identify External Exposure Points

Collect Credentials from
Unsecured sources

IoCs

Unusual access to IT/OT network

Opening of restricted Folders, Files
Access during off-hours

Logs detected on Supply chain
related systems

Port Scanning from suspicious IPs,
Request to Firewalls

Access to user desktops or email

Tactical Decisions Choices

1. Enforce Security
Monitoring Tools

2. Apply Least privilege
principles for Sensitive
Operational Data

3. Deploy Insider Threat
Detection Tools

4. Temporary
Suspension of At-Risk
Accounts

Scenario 2- Mining Materials Conveyer Belt Ops- Attacker Tactics

Weaponisation

Attackers Tactics

Backdooring Remote Access Tools

Insider develops scripts to pivot into IT/OT networks

Embed Malware in Legitimate Documents

Supply Chain Spoofing for OT Drivers

IoCs

Unexpected modifications or hash mismatches in legitimate tools

New/altered registry entries enabling silent startup or hidden services

outbound connections from remote desktop software detected

Unusual authentication logs outside of approved times

Tactical Decisions Choices

1. Isolate and Monitor Dual-Access Users (IT & OT)

2. Enable application whitelisting, block PowerShell/Bash scripting

3. Disable all non-essential remote tools, enforce MFA

4. Mandate signed firmware/driver uploads

Scenario 2- Mining Materials Conveyer Belt Ops- Attacker Tactics



Delivery

Attackers Tactics

Use of Compromised Internal Email Accounts for Malware Delivery

Physical Delivery of Malware via USB to OT Engineering Workstation

Abuse of Shared Devices and Network Folders

Deployment via Legitimate Software or Maintenance Tool

IoCs

Unusual email activity from privileged/internal accounts

New/altered registry entries enabling silent startup or hidden services

Unusual access/modification of shared folders by non-admin or off-shift users.

Unscheduled software/tool installation events in OT or IT asset logs.

Tactical Decisions Choices

1. Suspend compromised accounts and initiate forensic email tracing.

2. Enforce removable media policies and disable USB ports on OT assets.

3. Isolate and audit shared drives; apply access segmentation.

4. Suspend maintenance workflows pending integrity checks.

Scenario 2- Mining Materials Conveyor Belt Ops- Attacker Tactics



Exploitation

Attackers Tactics

Activate malware embedded in operator-used HMIs to interfere with conveyor logic

Exploit Default /Weak Authentication OT Systems

Abuse Scheduled Tasks on Engineering Workstations

Exploit Unpatched SCADA/HMI Vulnerabilities

IoCs

Unexpected changes in HMI, unathrorised logic cahngesin PLCs

Login attempts form unsual IPs

Unknown scheduled tasks appearing in Task Scheduler

Network scans targeting known OT software ports

Tactical Decisions Choices

1. Isolate Compromised OT Segments

2. Perform Endpoint Forensics on Engineering Workstations

3. Revoke or restrict elevated access rights for certain personnel

4. Start critical patches to address known vulnerabilities

Scenario 2- Mining Materials Conveyer Belt Ops- Attacker Tactics

Installation

Attackers Tactics

- Deploy Remote Access Tools (RATs) on Engineering Workstations
- Establish Persistence via Scheduled Tasks
- Side-load Malicious DLLs into OT Application Paths
- Create Unauthorized Local Admin Accounts on OT Machines

IoCs

- Unexpected changes in HMI, unathrorised logic cahngesin PLCs
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Tactical Decisions Choices

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Scenario 2- Mining Materials Conveyer Belt Ops- Attacker Tactics



Command & Control

Attackers Tactics

Deploy Remote Access Tools (RATs) on Engineering Workstations

Establish Persistence via Scheduled Tasks

Side-load Malicious DLLs into OT Application Paths

Create Unauthorized Local Admin Accounts on OT Machines

IoCs

Unexpected outbound connections from OT workstations

Unusual scheduled tasks not aligning with operational routines

Unusual DLLs found in HMI/SCADA software directories

Creation of unexpected user accounts with admin privileges

Tactical Decisions Choices

1. Segregate IT-OT Traffic via Firewall/Layer 3 ACLs

2. Authorize Temporary OT Downtime

3. Trigger Threat Management Playbooks

4. Monitor for beaconing patterns or anomalous outbound traffic

Scenario 2- Mining Materials Conveyor Belt Ops- Attacker Tactics



Objectives & Actions

Attackers Tactics

Sabotage Conveyor Belt Logic

Trigger False Safety Alarms

Data Exfiltration of Operational Logs

Manipulate Maintenance Schedule

IoCs

Unscheduled modifications to conveyor belt control code/logic blocks.

High frequency of alarms not matching real-world sensor inputs.

Unauthorized file movements from HMI/SCADA to file shares or USBs

Alterations in automated logs or updates made outside maintenance windows.

Tactical Decisions Choices

1. Immediate isolation of OT systems related to conveyer belt Ops

2. Start incident response with forensics

3. Should security teams treat this as an insider attack

4. To track attacker deploy deception techniques

De- Briefing Attack Scenario 2

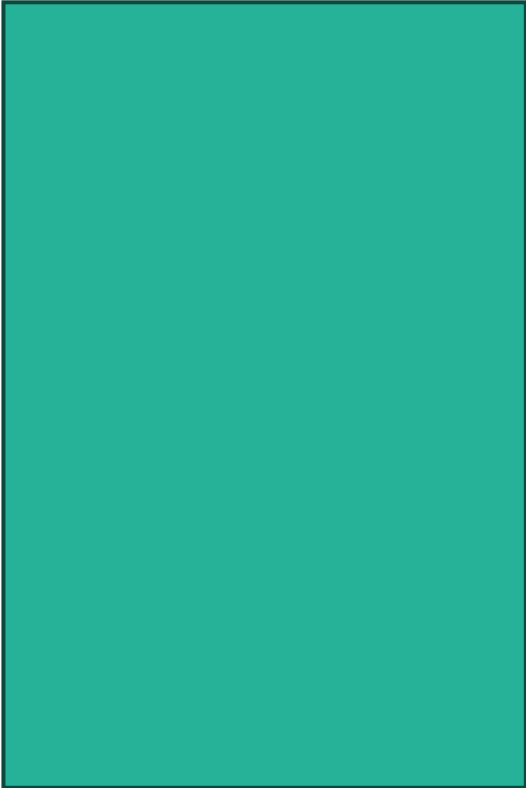
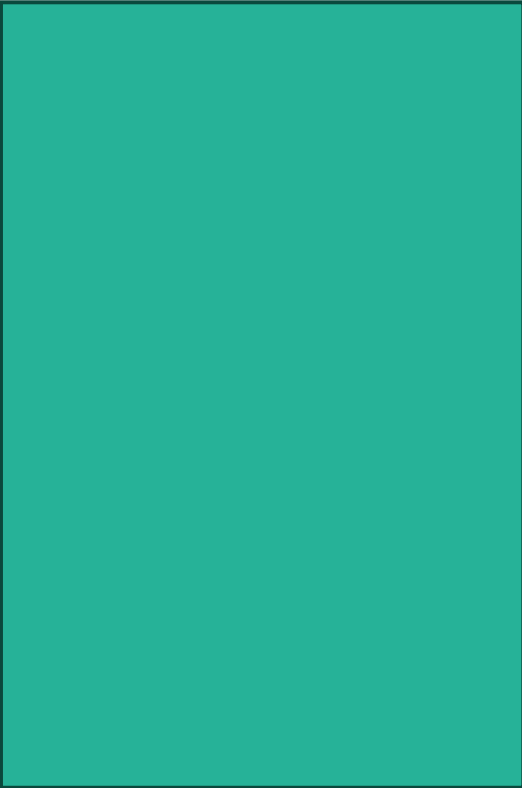
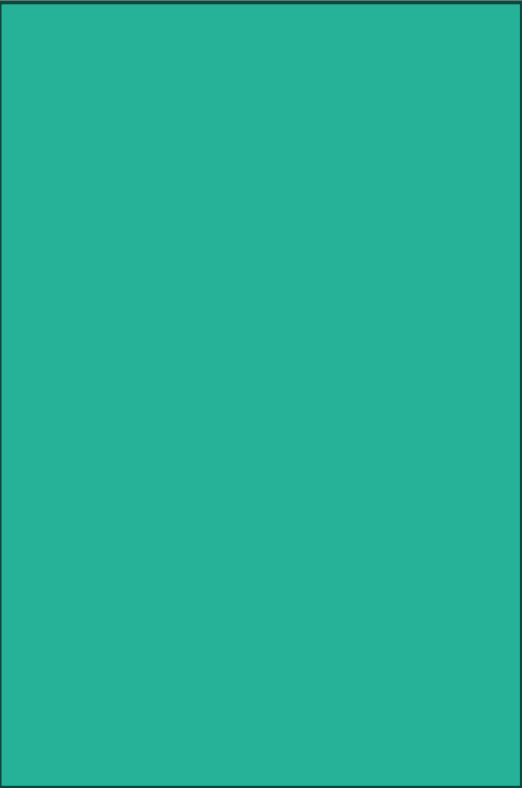
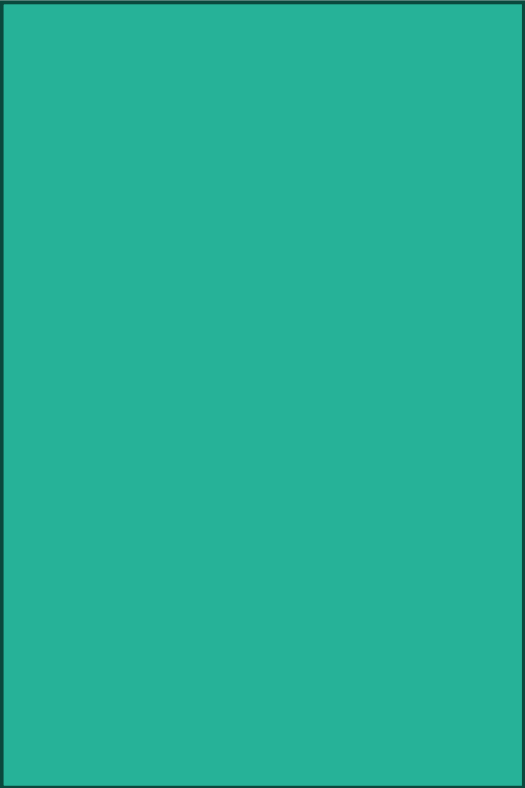
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Towards Resilient Mining Sector



Towards Resilient and Trustworthy Systems

What Measures should be taken to improve security posture ?

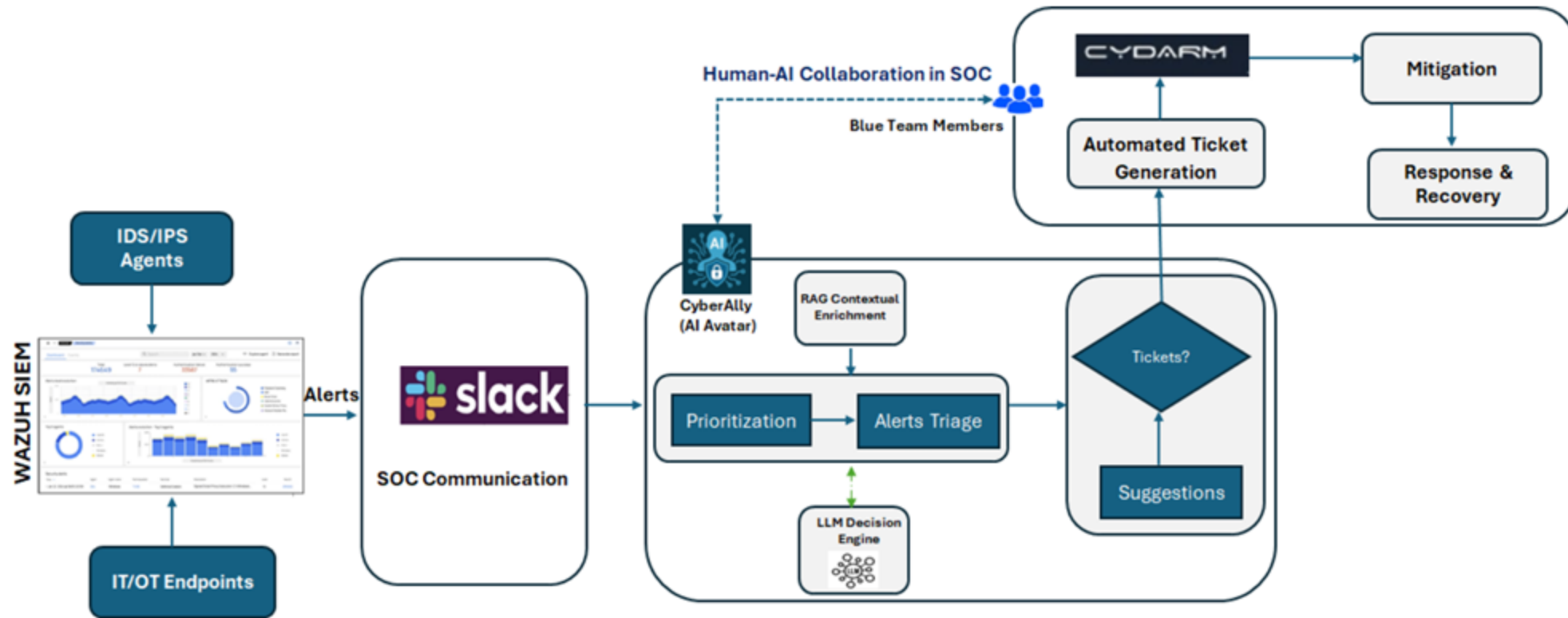
Identify	Protect	Detect	Respond	Recover
CI Assets Inventory Hardware + Software + People	Security Controls	OT/IT Data Ingestion	Cyber incidents coordination OT/IT	Disaster Recovery and Planning
OT/IT Governance	Secure by Design	OT/IT Threats Intelligence	Risks prioritisation	OT/IT Communication
OT/IT Risk Management	OT/IT Networks segmentation	Real Time Monitoring	Integrated IR at OT systems levels	Improvements
Business Processes Alignment	Patching and Vulnerabilities	OT/IT Protection Technologies		
Supply Chain Risk Management	Maintenance	Vulnerabilities Detection		

Why AI in SOC?

- Rising volume & complexity of threats
- Analyst fatigue & alert overload
- Need for real-time detection and response

AI Capability	SOC Impact
Threat Detection	Early identification of anomalies & breaches
Alert Triage Automation	Filters false positives; reduces response time
Incident Response Suggestions	Recommends best course of action
Threat Intelligence Correlation	Links global IOCs to local incidents
Behavioral Analysis	Detects insider threats, lateral movement

WAZUH SIEM



Cybersecurity Research Capabilities at ECU

At ECU's Centre for Securing Digital Futures, we lead innovation-driven cybersecurity research focused on enhancing Australia's critical infrastructure resilience.

Themes:

- **Securing Critical Infrastructure**
 - Maritime Port Systems (e.g., ACDC Project)
 - Advanced Manufacturing (Industry 5.0)
 - Water Systems & SCADA Networks
 - Distributed Energy Resources (DERs) and Energy Systems

AI driven innovation solutions

Digital Twins for Resilient Systems

